

Week 5 Task 2 Report

Topic: Identity & Access Management (IAM) and Cryptography Basics

Objective: To learn and implement IAM roles, policies, and cryptography concepts and document the practical steps in an organized way.

Part 1: Identity & Access Management (IAM)

Goal: To explore IAM policies, roles, and permissions in Azure, learn about Multi-Factor Authentication (MFA), and implement Role-Based Access Control (RBAC) by setting up roles and policies.

Implementation of IAM Roles and Policies in Azure

Step 1: Logging into Azure Portal

- Accessed the Azure Portal via <https://portal.azure.com>.
- Used my Azure credentials and authenticated using Multi-Factor Authentication (MFA).

Step 2: Setting up Role-Based Access Control (RBAC)

RBAC allows us to assign specific roles and permissions to users, ensuring that they have access only to the resources required for their job.

- Navigate to Subscriptions:
 - Opened the 'Subscriptions' tab and selected the subscription to configure RBAC.
- Assigned IAM Roles:
 - Under 'Access Control (IAM)' → Clicked on 'Add Role Assignment'.
- Created a Custom Role:
 - Defined a new role with specific permissions (e.g., allowing read-only access).
- Configured Multi-Factor Authentication (MFA):
 - Enabled MFA for selected users for additional security.

Below is a summary of the roles, users, and permissions configured:

User/Group	Assigned Role	Scope	Purpose
Team Member 1	Contributor	Subscription	Manage resources

			without IAM access
Team Member 2	Reader	Specific Resource Group	Read-only access
Admin	Owner	Entire Subscription	Full access to manage all aspects

Part 2: Cryptography Basics

Goal: To understand and implement core cryptography concepts, including encryption, decryption, hashing, and RSA key pair generation.

AES-256 Encryption and Decryption Using OpenSSL

- Created a Sample Text File using: `echo 'Sample text' > plaintext.txt`
- Encrypted the File with AES-256: `openssl enc -aes-256-cbc -salt -in plaintext.txt -out encrypted.txt`
- Decrypted the File: `openssl enc -aes-256-cbc -d -in encrypted.txt -out decrypted.txt`

RSA Key Pair Generation

- Generated RSA Private Key: `openssl genrsa -out private_key.pem 2048`
- Extracted the Public Key: `openssl rsa -in private_key.pem -pubout -out public_key.pem`
- Encrypted Data using Public Key and Decrypted with Private Key.

Hashing with SHA-256

- Calculated SHA-256 Hash: `openssl dgst -sha256 plaintext.txt`
- Verified File Integrity by Comparing Hash Values.

Final Conclusion:

By completing these tasks, I gained hands-on experience in configuring IAM roles, policies, and MFA in Azure, and implementing AES-256 encryption, RSA key pair generation, and SHA-256 hashing using OpenSSL.